

Smart Metro Ticket Booking System

Project Design Phase

Date	
Team ID	
Project Name	Smart Metro Ticket Booking System
Maximum Marks	4 Marks

2. Proposed Solution

The proposed solution is a ServiceNow-based Smart Metro Ticket Booking system that lets passengers look up routes, check fares, pay digitally, and get a QR ticket within minutes — while giving Station Managers, Metro Operations, and IT Admins the tools to manage stations, track performance, and configure the platform without relying on counters or custom code.

2.1 Solution Overview

At its heart, the solution turns the entire booking-to-travel journey digital. A passenger picks a source and destination from a searchable catalog form, sees an itemized fare instantly (with passenger-type discounts applied automatically), pays via UPI, card, or wallet, and gets a unique, single-use QR ticket delivered by app notification, email, and WhatsApp/SMS. That same QR code is checked at the station gate in real time, with every step logged for audit, reporting, and exception handling.

2.2 End-to-End Passenger Journey

Proposed Solution — Passenger Booking Journey

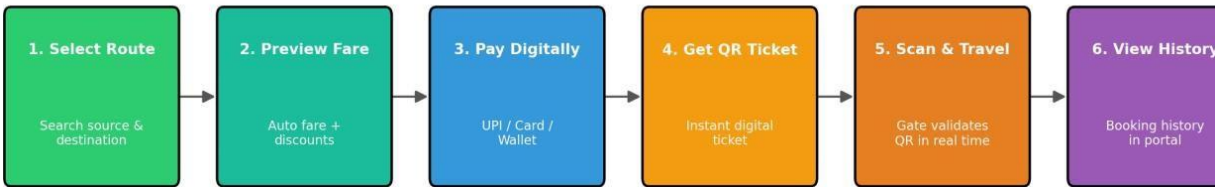


Figure — Passenger journey from route selection to travel

2.3 Key Capabilities

- A searchable Service Catalog booking form with source/destination validation against the Station Master.
- Real-time, itemized fare calculation with automatic Student/Senior Citizen discounts.
- Digital payments via UPI, debit/credit card, and mobile wallets, with a clear retry path on failure.
- Instant, single-use QR ticket generation tied to the passenger's booking record.
- Multi-channel delivery of tickets and alerts — in-app, email, and WhatsApp/SMS.
- Booking history and ticket re-download through the Service Portal.
- Station Manager tools for master-data maintenance and real-time gate validation.
- Metro Operations dashboards for ticket volume, revenue split, and peak-time analytics.
- IT Admin configuration of catalog variables, fare rules, and role-based access — no code changes needed.

2.4 Value Proposition

Persona	Value Delivered
Passengers	Book and travel in under 5 steps, with no queue or cash needed.
Station Managers	Faster, verifiable gate checks and less manual data upkeep.
Metro Operations	Data-driven scheduling and revenue tracking, updated in near real time.

Persona	Value Delivered
IT Admins	Faster turnaround on fare and configuration changes, with full traceability.